Anoushkrit Goel

Applied Scientist | **1x Entrepreneur** | **Spatial AI (3D and 2D)** | **Medical AI** | **AI Researcher**

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Experience

Research Engineer 2, *Attentive.AI* – Gurgaon, IN current Jun 2025 – Present

- Architected a vector path tokenization framework (PDF → SVG) as a pretraining objective, for zero-shot symbol detection and vector path annotation, increasing SOTA by 7%, avg macro-F1. ***[PDF, SVG, Doc AI]***
- Trained and shipped **DEIMv2** (**DINOv3** backbone) for multi-class vector paths segmentation and geometry dimensions estimation. Reducing total annotation time (TAT) for HVAC to 35% of original time.
- Designed, built and deployed a novel OCR for blueprints, **ArchGlyphOCR**, a text-region proposal + prediction pipeline for dense and high resolution plans, outperforming **Google DocAI** and **Gemini 3.1 Pro** on callout extraction. . Eliminating API costs, rework and longer wait-times for DIY tool. ***[OCR, VLM]***

Researcher, *IIT Mandi* – Mandi, IN Feb 2023 – Jun 2025

- Designed foundational models for 3D streamlines using **PointGPT** style architectures, with novel tokenization, dual-masking pretraining, multi-embedding for WM tract segmentation.
- Built and deployed **SlicerTracto**, an open-source end-to-end 3D Slicer extension for brain MRI tractography (reconstruction, tracking, segmentation, visualization); deployed at **PGI Chandigarh Hospital** with options to infer from local / on-prem / cloud GPUs
- Published **5 papers**: ICPR'24 (TractoEmbed[1], TractRL-former[2]), ISBI'25 (TractoGPT)[3], ISBI'26 (TrackletGPT[4], TractRLFusion)[5], **outperformed Harvard Medical School** (TractCloud) and **SCIL Canada** (FIESTA). GPT, BERT, RL-transfomers on 3D point cloud spatial data.

Lead Data Scientist, *StatusNeo* – Gurgaon, IN Aug 2022 – Mar 2023

- Led Cortex AI team for **Reliance Jio** (BrainOS) spanning across Jio's verticals, where we built framework-agnostic ML library via *protobuf* schemas and *ONNX*; mapped model ontology, parsed 6 sklearn models and PyTorch libraries into a single cross-framework schema (sklearn ↔ PyTorch ↔ TF). ***[Core AI]***

Senior Data Scientist, *Gamezop* – Gurgaon, IN Jun 2021 – Jun 2022

- Led **ClickVision**: MDP-formulated deposit prediction over clickstream trajectories using RL transition matrices; reduced wasted offers and churn by **62%**, improving deposit conversion; downstream UI changes were shipped based on model findings. ***[Behavioral RL]***

Founder & CEO, *Tensr.AI* – Noida, IN Feb 2020 – Jun 2021

- Built **pyCardiograms** (LSTM-based SCG waveform prediction, wave annotation across ECG/PPG/SCG). Scaled to 15 members; **DST DPIIT** certified; incubated at **SIIC IIT Kanpur**, HTIC **IIT Madras**, **IIT Mandi Catalyst**; ranked **#3** Himalayan Startup Trek 2019. ***[Biomedical Signal]***
- Designing, prototyping, and building our proprietary health wearable, **Teresa Band**, to capture the biomedical signals.

Data Scientist, *xtLytics LLC* – Noida, IN Jan 2019 – Jan 2020

- **Invoice OCR**: GCP Vision + Tesseract + Faster-RCNN to extract and structure loosely-tabular document content into .csv/.xlsx via geometric bounding-box heuristics. ***[Document OCR]***
- **CADtnet**: Developed an edge TFLite model (VGG-19, MobileNet, ResNet), for a mobile app to perform breast cancer triage on ultrasound images, deployed in LMICs (Low-Middle Income Countries). ***[Biomedical Image]***

Education

Indian Institute of Technology Mandi, M.Tech(R) *Intelligent Systems* **9.47/10** Feb 2023 – Nov 2025
Shiv Nadar University, B.Tech Electronics Engineering | Minor: Design | 7.61/10 Jul 2015 – May 2019

Tech Stack, Skills & Certifications

Languages: *Python*, MATLAB, C, C#, ProtoBuf, LaTeX **Cert:** **DL Specialization** [WXBN3CDY3J33]
Document AI: PDF Parsing, SVG / Vector Paths (bézier, polylines, fills), OCR (Tesseract, GCP Vision, VLMs), Document Layout Analysis, CadTransformer, VecFormer, Sympoint, ArchCAD, **DEIMv2**, **DINOv3**, GLM
Deep Learning: PyTorch, HuggingFace, ViT/DINO, Transformers (GPT, BERT), PointGPT, PointBERT, Open3D, ONNX, TensorFlow/Keras, scikit-learn, OpenCV, GeoPandas, scipy, Tensorboard
For *LLMs & GenAI*: LangChain, LangGraph, RAG, GraphRAG, Multimodal LLMs (Text + Image + 3D), AutoGen, DSPy, ollama, AWS, GCP, PyMuPDF
***Others*:** Simulink, 3D Slicer, MRtrix3, FSL, FreeSurfer, Unity ML Agents, Adobe Illustrator, Verilog